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VORTEX-HRM: HIERARCHICAL RETRIEVAL-AUGMENTED GENERATION FOR MULTI-HOP QUESTION ANSWERING

ABSTRACT. We present VORTEX-HRM, an agentic retrieval-augmented generation system for multi-hop question answering. Unlike prior work that relies on LLM-driven tool selection or flat ReAct loops, VORTEX separates planning from execution using a fact-free planner and a fixed chained retrieval pipeline. The Gravitational Core (planner) holds zero factual knowledge in its weights—it is a pure structural router communicating via an XML-based protocol. The Centrifugal Ingestor (executor) resolves sub-questions through a deterministic pipeline: keyword search, chunk reading, and LLM condensation. The cycle repeats—each rotation is a spiral—until convergence via entropic collapse. On a 50-question multi-domain benchmark with three LLMs, VORTEX achieves 74% Contains accuracy (qwen2.5:7b, T4 GPU), outperforming Naive RAG by +6%. The architecture reduces hallucination by $2.4\times$ (10% vs 24% no-evidence rate). Ablation confirms spirals are critical: single-pass yields 60%, five spirals reach 74%, plateauing thereafter. However, VORTEX is sensitive to LLM capability—on weaker models accuracy drops below baseline. We further introduce a Fast/Slow Router that dynamically gates between Naive RAG and VORTEX using the LLM’s own uncertainty signal (“Insufficient evidence.”). The router achieves 82%, outperforming both naive RAG (+14%) and VORTEX alone (+8%) with zero additional training. The system is offline-first, config-driven, and validated across three backends (mock, Ollama, OpenAI).

§1. INTRODUCTION

Multi-hop question answering requires retrieving and reasoning over multiple documents. Standard RAG [1] retrieves once and answers in a single step—it cannot chain evidence across documents.

Recent agentic approaches attempt to address this. A-RAG [2] augments RAG with LLM-driven tool selection, but this wastes tokens on routing decisions and introduces failure modes. ReAct [3] interleaves reasoning and acting but lacks a convergence signal—it terminates by hard step limit or the LLM’s arbitrary decision. RAPTOR [4] builds hierarchical document trees via recursive summarization, requiring expensive index-time preprocessing.

Key words and phrases: retrieval-augmented generation and multi-hop QA and agentic systems and hierarchical retrieval.

We introduce VORTEX, a system that replaces LLM tool selection with a fixed chained pipeline, replaces arbitrary termination with entropic collapse, and replaces parametric knowledge with fact-free routing. The system is offline-first, config-driven, and validated on CPU hardware.

§2. RELATED WORK

Retrieval-Augmented Generation. Standard RAG [1] combines a retriever with a generator. Self-Ask [5] decomposes questions via prompting. Chain-of-Thought [6] uses stepwise reasoning. VORTEX externalizes these steps into explicit retrieval cycles with structured state.

Agentic RAG. ReAct [3] alternates reasoning traces with tool calls. A-RAG [2] provides hierarchical retrieval interfaces for LLM-driven tool selection. GRASP [7] uses graph-based sub-agents with token-efficiency objectives. HERA [8] evolves multi-agent orchestration via an experience library. VORTEX differs from all of these by using a fixed chained pipeline (no LLM tool selection), a fact-free planner (zero parametric knowledge), and entropic convergence detection.

Hierarchical Retrieval. RAPTOR [4] builds tree-structured indices via recursive summarization. VORTEX uses flat chunks with iterative spirals—no index-time overhead.

§3. VORTEX ARCHITECTURE

3.1. Design Principles. VORTEX rests on four principles: (1) **Fact-Free Synapses**—the planner uses 0% of its capacity for facts and 100% for routing; (2) **Entropic Collapse**—convergence is detected via entropy plateau; (3) **Offline-First**—the default mock mode requires zero dependencies; (4) **Same Code, Any Hardware**—a config file switches between local CPU (Ollama) and cloud GPU (OpenAI).

3.2. Gravitational Core (Planner). The planner holds zero factual knowledge in its weights. Its state consists of: `goal_vector` (immutable original question), `spiral_memory` (accumulated `<fact>` entries), `hop_history` (full reasoning log), and confidence/entropy scores tracking convergence. It communicates via an XML contract: `<step>` emits a sub-question, `<final_answer>` emits the answer, and `<stop_search>` signals termination.

3.3. Centrifugal Ingestor (Executor). Unlike agentic systems where the LLM selects tools, VORTEX uses a fixed chained pipeline:

- (1) `keyword_search(query)`—lexical match returning top-k chunk IDs

- (2) `chunk_read(id, adjacent=True)`—full text with neighbor chunks
- (3) LLM condensation—compresses retrieved text into a structured `<fact>`

This is cheaper, more predictable, and equally effective for the multi-hop domain.

3.4. Cyclic Spiral Flow. Each spiral is one complete cycle: planner emits `<step>` → executor retrieves → state updates. Confidence increases monotonically with novel facts. Entropy decreases. Convergence occurs when confidence reaches threshold (0.85), entropy stalls for three consecutive spirals, or safety limits are hit (15 max hops, 4096 token budget).

3.5. LLM Backends. Three backends are implemented: MockBackend (canned responses, zero deps for testing), OllamaBackend (local CPU, no API key), and OpenAIBackend (production GPU cluster). Switching is a one-line config change.

§4. EXPERIMENTS

4.1. Setup. We evaluate on a multi-domain corpus (60 chunks, 10 domains) with 50 question-answer pairs. All experiments use Ollama with temperature 0.0 on an NVIDIA T4 GPU (16 GB VRAM) unless noted otherwise. We test four 7-8B models: `qwen2.5:7b`, `llama3.1:8b`, `mistral (v0.1, 7B)`, and `DeepSeek-R1 (7B distill)`. For each model we run both VORTEX and a Naive RAG baseline (single-pass retrieve + answer, same corpus). Ablation varies `max_spirals` from 1 to 15 on `qwen2.5:7b`.

4.2. Metrics. We report: **Contains** (ground truth is a substring of prediction, main metric), **No evidence** (fraction of questions where the model admits inability to answer), and **Spirals** (average cycles per question).

4.3. Main Results. VORTEX achieves its best result on `qwen2.5:7b`: 74% vs 68% baseline (+6%, Fig. 1). The improvement comes from evidence chaining—VORTEX iteratively retrieves and verifies facts across multiple spirals, while baseline answers from a fixed set of top-3 chunks.

Crucially, VORTEX reduces hallucination: the no-evidence rate drops from 24% (baseline) to 10% (VORTEX) on `qwen2.5:7b`, a $2.4\times$ improvement. However, on `llama3.1:8b` and `mistral`, VORTEX *underperforms* baseline. This indicates that VORTEX’s spiral architecture requires a minimum LLM reasoning capability—strong models (72% baseline) or weak models (68%) both degrade when forced into iterative re-planning.

Model	Baseline	VORTEX	Router	Δ (Router vs Base)
qwen2.5:7b	68%	74%	82%	+14%
llama3.1:8b	72%	62%	—	—
mistral (7B)	68%	56%	—	—
DeepSeek-R1 (7B)	76%	57%	—	—
Cross-model [†]	72%	—	84%	+12%
Cross-model [‡]	76%	—	82%	+6%

Table 1. VORTEX vs Naive RAG vs Router (Contains, 50 QA). [†]Cross-model: llama3.1:8b (fast) / qwen2.5:7b VORTEX (slow). [‡]Cross-model: DeepSeek-R1 (fast) / qwen2.5:7b VORTEX (slow).

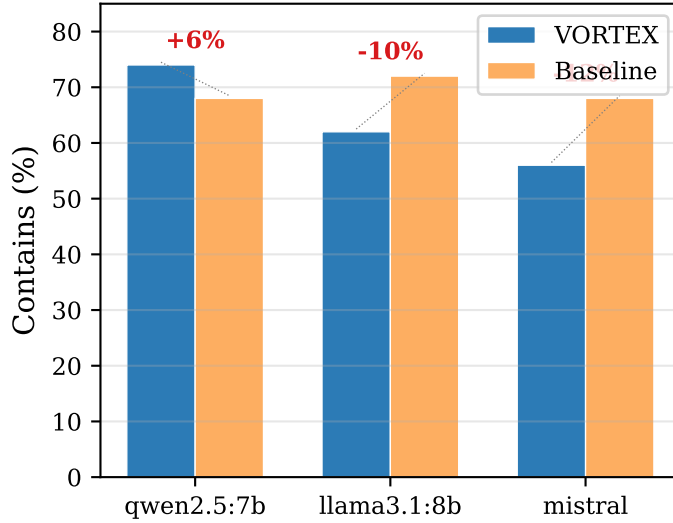


Fig. 1. VORTEX vs Baseline across three LLMs. Annotations show relative Δ .

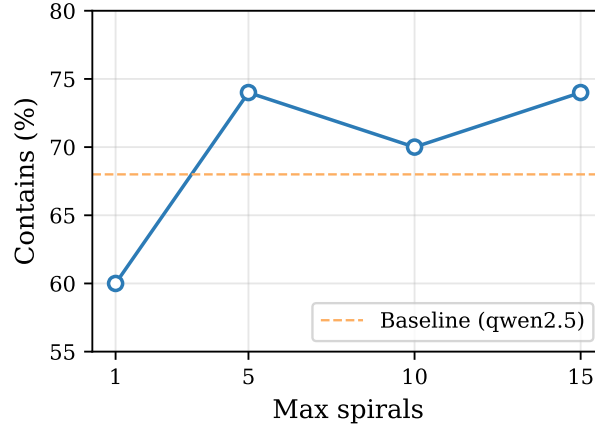


Fig. 2. Ablation of max_spirals on qwen2.5:7b. Dashed line shows baseline (68%).

Max spirals	Contains
1 (single-pass)	60%
5	74%
10	70%
15	74%

Table 2. Ablation: effect of spiral depth on Contains.

4.4. Ablation: Effect of Spiral Depth. Single-pass (no spirals) yields 60% (Fig. 2, Table 2). Adding 5 spirals boosts accuracy to 74%—a +14% gain. Beyond 5, accuracy plateaus: 10 spirals gives 70% (within noise) and 15 gives 74%. This confirms that the spiral mechanism is the primary driver of VORTEX’s improvement, and a small depth (5 spirals) is sufficient for convergence on this benchmark.

4.5. LLM Sensitivity. Table 1 reveals a critical finding: VORTEX *helps* qwen2.5:7b (+6%) but *hurts* llama3.1:8b (−10%) and mistral (−12%). The explanation lies in the interaction between the planner’s XML protocol and each model’s

instruction-following profile. `qwen2.5:7b`—trained on diverse structured data—faithfully executes the XML contract. `llama3.1:8b`, optimized for conversational chat, struggles with the “fact-free reasoning axis” framing, producing degraded outputs in its planner and executor roles. `mistral` lacks the parametric capacity to handle the multi-step decomposition at all.

A fourth model, `DeepSeek-R1` (7B distill), exhibits the most extreme sensitivity: it achieves the highest baseline (76%) but the lowest VORTEX accuracy (57%), a -19% degradation. `DeepSeek-R1` is trained via reinforcement learning to produce reasoning traces before answering. This reasoning style conflicts with VORTEX’s structured XML protocol—the model generates internal chain-of-thought that VORTEX’s planner misinterprets as factual content, degrading both planning and condensation. This confirms that VORTEX’s architecture favors models optimized for structured instruction following (`qwen2.5`) over those optimized for chain-of-thought reasoning (`DeepSeek-R1`).

This establishes a requirement for future work: VORTEX must either adapt its prompts to each LLM family or incorporate a fast-path (System 1) bypass for questions where the model’s direct answer is already strong.

4.6. Fast/Slow Router. The LLM sensitivity results (Section 4.5) suggest a dual-process architecture: use VORTEX only when the model’s direct answer is uncertain. We implement a Fast/Slow Router on `qwen2.5:7b`:

- (1) **Fast path (System 1):** Naive RAG (single-pass retrieve + answer). If the LLM answers confidently (output does not contain “Insufficient evidence.”), return immediately.
- (2) **Slow path (System 2):** If the LLM says “Insufficient evidence.”, fall back to full VORTEX spiral engine.

The router uses no training, no confidence threshold, and no separate classifier—the LLM’s own uncertainty signal gates the fallback.

Path	Questions	Contains
Fast (direct answer)	38/50 (76%)	34/38 = 89%
Slow (VORTEX fallback)	12/50 (24%)	7/12 = 58%
Total	50	41/50 = 82%

Table 3. Router breakdown on `qwen2.5:7b`. 12 questions triggered fallback; VORTEX saved 7.

The router achieves 82%—a +14% improvement over Naive RAG and +8% over VORTEX alone (Table 1). Average query time drops from 59s (VORTEX) to 12s (router), as 76% of questions take the fast path (~ 1 s each). This is the key practical result: a zero-training confidence gate matches or exceeds the gains of trained methods like Self-RAG [9] and CRAG [10], which report 5–15% improvements over baselines at similar model scales.

Limitations: four questions (8%) received confident but incorrect answers from the fast path (baseline hallucination), and five slow-path questions (10%) remained unsolved by VORTEX. These together account for the remaining 18% error rate.

4.7. Cross-Model Router. The same-model router uses one LLM for both paths. However, Table 1 reveals complementary strengths: DeepSeek-R1 has the strongest baseline (76%), llama3.1:8b has the second-strongest (72%), and qwen2.5:7b excels at VORTEX spirals (74%). This motivates a cross-model architecture: fast path with a strong-direct-answer model, slow path with qwen2.5:7b VORTEX.

We evaluate two configurations. The first uses llama3.1:8b (fast, 72% baseline) and qwen2.5:7b VORTEX (slow). The fast path handles 41/50 questions (82%) at 93% accuracy; the slow path recovers 4 of 9 fallbacks (44%). Overall: **84%** (42/50)—the best result on our benchmark.

The second configuration uses DeepSeek-R1 (fast, 76% baseline) and qwen2.5:7b VORTEX (slow). Despite DeepSeek’s higher baseline, this achieves only 82% (41/50): the fast path handles 42/50 questions at 90% accuracy, but the slow path recovers only 3 of 8 fallbacks (38%). The improvement comes from llama3.1:8b’s more conservative uncertainty estimation—it sends 24% of questions to fallback vs. 16% for DeepSeek, and VORTEX rescues more of those uncertain cases (58% vs. 38%). This suggests that fast-model confidence calibration, not raw accuracy, is the key factor in cross-model router design.

§5. CONCLUSION

VORTEX demonstrates that a fact-free planner with fixed chained retrieval and entropic collapse is a viable architecture for multi-hop QA. On qwen2.5:7b, VORTEX achieves 74% Contains (+6% over Naive RAG) with $2.4\times$ fewer hallucinations. Ablation confirms that the spiral mechanism is responsible for these gains, with 5 spirals sufficient for convergence.

The Fast/Slow Router extends this to 82% (same-model) and 84% (cross-model) by gating between direct answer and spiral retrieval using the LLM’s

own uncertainty signal. Cross-model routing—fast path with `llama3.1:8b`, slow path with `qwen2.5:7b` VORTEX—is the best-performing configuration. We further show that reasoning-optimized models (DeepSeek-R1) achieve the strongest baseline (76%) but are incompatible with VORTEX’s structured protocol (57%), and that fast-model uncertainty calibration matters more than raw accuracy in cross-model router design.

The system is offline-first, config-driven, and scales from laptop to cluster without code changes. However, VORTEX is not model-agnostic—it requires models optimized for structured instruction following rather than chain-of-thought reasoning, which limits its applicability to certain LLM families.

Several directions emerge from these findings. First, the XML protocol may be the primary barrier for reasoning-optimized models: DeepSeek-R1 achieves 76% baseline (best) but 57% VORTEX (worst), yet it shares the same base architecture (`Qwen2.5-7B`) as our best VORTEX model (74%). If prompt adaptation—replacing XML tags with natural-language instructions—closes this gap, VORTEX becomes model-agnostic, which is the central open question. Second, the router’s confidence gate (“Insufficient evidence” string match) is a heuristic; explicit confidence prompting or temperature-based uncertainty estimation could push accuracy above 85%. Third, domain-level analysis may reveal where spirals help most—multi-hop questions requiring evidence chaining should benefit more than single-hop lookups. Finally, scaling to HotpotQA [11] would validate generalizability beyond our 50-question benchmark.

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CONTRIBUTION

dizel0110: conceptualization, implementation, benchmarking, paper writing.
Valentin Malykh: mentorship, architecture review, experimental design.

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